

Smart Hydroponic Garden System





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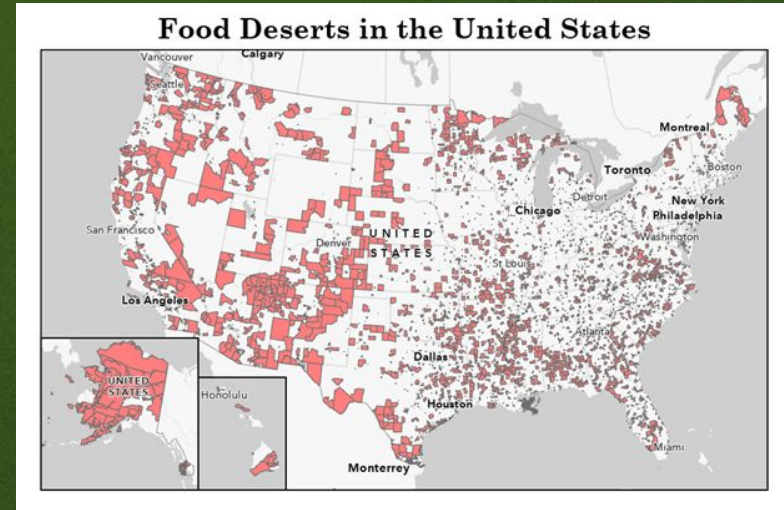
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(01)

Project Overview

Growing Food in a City That Has No Room for Gardens

- 23.5M Americans live in urban food deserts [1]
- 2/3 of global population growth (2025–2050) will be in cities [2]
- 35% of U.S. households want to grow their own food, but say they lack the time and the know-how [3]



Food Desert Map according to US census (2019)

(02)

Problem & Solution

Problem & Solution

Problem

- Existing products (*AeroGarden*, *Lettuce Grow*) run on fixed timers, no real sensor feedback
- Commercial systems (*Hydrofarm*) provide automation but expensive and complex
- Gap: no affordable system gives home users real-time visibility into plant conditions

Solution

- Real-time monitoring of water temperature, humidity, pressure and pH
- Raspberry Pi 5 Controller, Sensor: BME280, BME680 and Atlas Scientific pH
- User Interface with threshold-based alerts when conditions go out of range

CMPE-195A

Group 18

Spring 2026

(03)

Demo

System Overview: A Functional End-to-end pipeline

SENSOR

- Temperature
 - Humidity
 - pH/EC
- > Sensors collect data
From hydroponic system
and send to backend

Sensor Data
(HTTP/JSON)

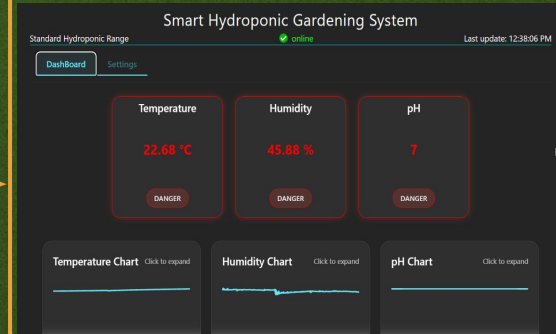
API (Backend)

Fast API:

- Handle request
- Process data
- Store it in database
- Return response

API Response
(JSON)

DASHBOARD



Frontend dashboard visualize
real-time and historical data
with charts and alerts



Collect

Sensors measure
real-world conditions



Transmit

Data is sent to
the backend via API



Visualize

Data is displayed
as insights on the dashboard

Demo

- Frontend UI Walkthrough: (POC) UI Demo For Smart Hydroponic Garden System
- API call Testing: (POC) Backend API Requests Demo

(04)

Evaluation Results

Evaluation Results

Metrics	Results
Temperature accuracy	$\pm 0.5^{\circ}\text{C}$ (repeatability $\pm 0.2^{\circ}\text{C}$)
Humidity accuracy	$\pm 2 - 5\%$ RH (repeatability $\pm 1\%$ RH)
pH Accuracy	Not delivered yet
Alert Latency	$< 30\text{s}$ from threshold breach to UI
API response time	$< 200\text{ms}$ at 95th percentile

(05)

Key Learnings

What we learned so far?

- **Open-loop system:** alerting before fully automation forced us to design the system with clean logic
- **ngrok offers team development:** exposing Pi-hosted backend let frontend work occur off-network

What we'd do differently?

- Start hardware-in-the-loop testing earlier
- pH sensor delayed delivery pushed simulated data further into the cycle we wanted

(06)

Conclusion

Impact solution:

- Affordable
- Sensor-driven hydroponic monitoring system for urban home-grower

Objectives for CMPE-195B:

- Closed-loop system with pump, light and dosing control
- Cloud-hosted frontend, real-time streaming via WebSockets. Alert history & acknowledgement.



Work Cited

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[Accessed: Mar. 13, 2026].

[3] National Gardening Association, "National Gardening Survey 2023," National Gardening Association, Williston, VT, 2023. [Online]. Available:

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**Thanks for your
attention!**